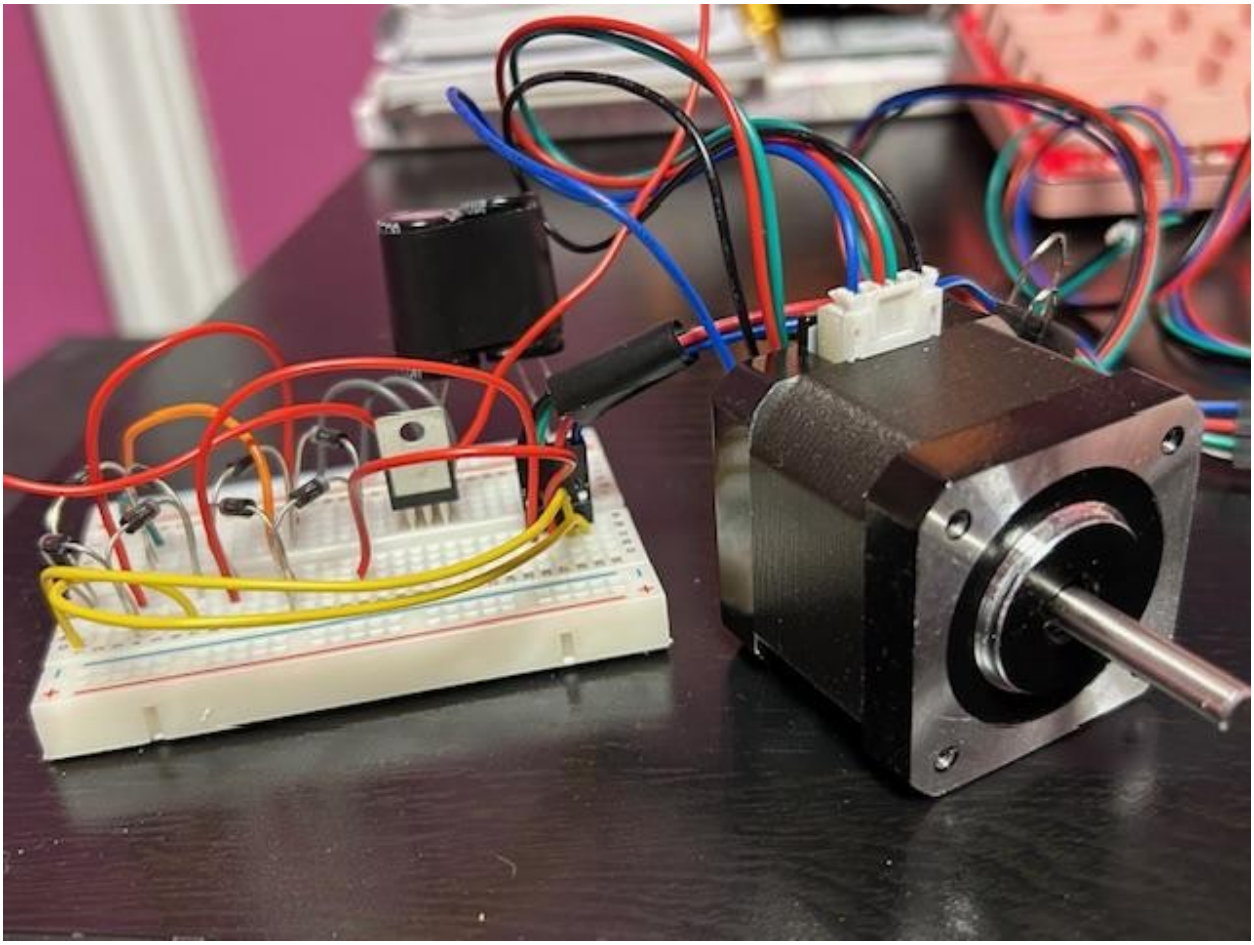
Hand Crank Generator Documentation Summary and Notes

Lina Habib



Hand Crank Generator

Objective:

Create a hand crank powered generator that outputs a steady 5V DC current to be used as a portable phone charger.

Design Process - organized by sequence of considerations

1. **Crank**

- Uses Nema-17 Stepper Motor; full 360 degree range of motion.

2. **Motor Output**

- Produces alternating current to be converted to direct current.
- To be resolved using a ***full bridge rectifier***.

- Current - based on Nema-17 model
- Higher RPM = Higher Voltage
- To be resolved using a ***voltage regulator***
- Voltage may also fluctuate due to inconsistent cranking speeds
- To be resolved using a ***capacitor***

3. **Nema-17 Model 17HS15-1504S-X1**

- Maximum Current Output: 1.50A / phase
- Maximum Voltage: Depends on load and resistance. Typically 12V to 15V with vigorous cranking.

4. **Circuit components**

- Diodes (x8) - Used to create full bridge rectifiers converting AC to DC by creating a one way path for current.
    - The 2-phase motor contains 2 coils, each with 2 wires connected to one of 2 full bridge rectifiers each composed of 4 diodes in a diamond shape to direct current. See image below.

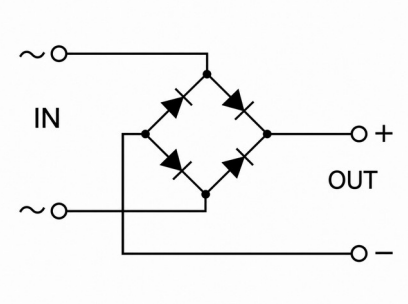
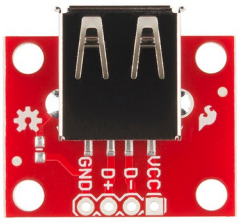


Figure 1. Full bridge rectifier consisting of 4 diodes. Direction of current is shown.

- Voltage Regulator (x1) - Restricts voltage to 5V (Voltage required to charge a regular phone)
- Capacitors (x2) - Stabilizes voltage at 5V. Stores and releases electrical energy to deliver steady, flat direct current.
    - Opted for 16V 3300μF capacitors, which can safely handle 5V and also have a large storage capacity for absorbing spikes in voltage.
- Breadboard (x1) - to connect the circuit components.
- Jumper wires
- Additional tool: Multimeter - for ensuring desired output voltage.

5. **Charger**

- Connect to a USB-A breakout board shown below to plug in lightning cable charging wire.

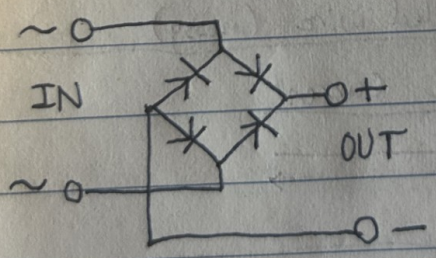


https://learn.sparkfun.com/tutorials/usb-type-a-female-breakout-hookup-guide/all

Preliminary Research Documentation - Unrevised

HAND CRANK GENERATOR

- Crank → Stapper motor?
  ↳ Generates AC
* Convert into DC
  ↳ Use a full bridge rectifier
- Full Bridge Rectifier
  ↳ 4 diodes (like diff waves)
* If stepper motor has 2 phases on each coil, you need 2 FBR



(rectifies sin waves into DC)

- 5v regulator
  ↳ combine DC lines in series before running it through regulator
  ↳ Regulator ensures we cant spin crank too fast and send too high voltage

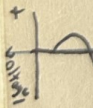
- Smoothing Capacitors
  ↳ consistent 5V
- PCB to connect to stepper motor
  not the same plug as that white board
- Bread board

LED Note: Long leg is positive - connect to positive.

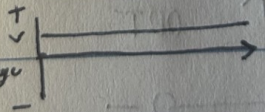
Resistor has no associated polarity

- Wires - BUY SOLID CORE WIRE
  not stranded wire (messy)

AC Current

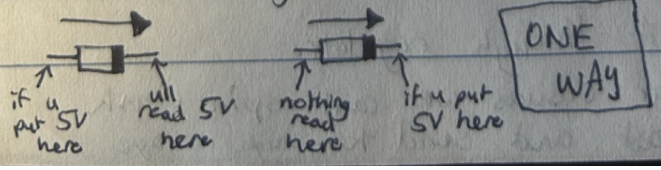


DC Current



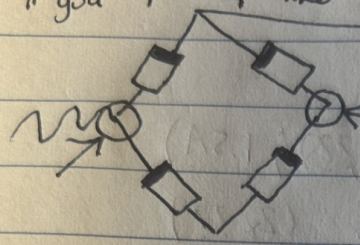
Rectifier notes

- Diodes - one way channels for electricity



FBR contd

if you put 4 like this...

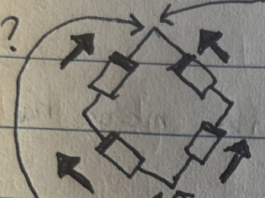


→ comes out DC, flat up here

put any AC, squiggly voltage here

it all pulls up at the top (POWER)

WHY?



electricity can't go against diodes

leaving none here (GROUND)

WHICH means... THIS IS NEG

THIS IS POS

electricity flows from neg to pos.

- Devices

- Multimeter – measures voltage
  - ↳ test 5V regulator
- Oscilloscope

Materials

- ✓ Stepper motor
- ✓ 8× diodes
- 5V regulator (linear, 5V, 1.5A)
- Smoothing capacitors (2x ??)
- ✓ Breadboard ≒14≒
- ✓ Jumper wires 10x unless motor has more
- Led bulb for insurance
- PCB - motor has female DuPoint connect

Budget

- Motor (w/o tax) • Amazon — $16.95
- Sayal Electronics — $41 *about

